Cisco Identity Services Engine Upgrade Guide, Release 3.3

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CHAPTER 1

Upgrade Cisco ISE

Cisco ISE upgrade overview

pxGrid Version 2.0, which is based on WebSockets, was introduced in Cisco ISE Release 2.4. We recommend that you plan and upgrade your other systems to pxGrid 2.0-compliant versions in order to prevent potential disruptions, if any, to integrations.

This document describes how to upgrade your Cisco ISE software on Cisco ISE appliances and virtual machines (VMs) to Release 3.3. (See the section "What is new in Cisco ISE Release 3.3" in the *Release Notes for Cisco Identity Services Engine, Release 3.3*.)

Upgrade planning should incorporate the estimated time required for each upgrade step to minimize service disruption. Deployments with multiple Policy Service Nodes (PSNs) grouped together typically experience no downtime during upgrades, whereas standalone or single PSN deployments may incur downtime.

Caution Standalone deployments are at higher risk of service interruption during upgrade. Plan maintenance windows accordingly and consider backup strategies to mitigate impact.

Note When upgrading to Cisco ISE release 3.2 and later, Root CA regeneration happens automatically in the upgrade process. Thus, post-upgrade Root CA regeneration is not required.

Different types of deployment

Cisco ISE deployments can be categorized as follows:

- **Standalone Node**: A single node running all personas (Policy Administration Node, Policy Service Node, and Monitoring and Troubleshooting).
- **Multi-Node Deployment**: A distributed deployment with dedicated nodes for different personas.

Differences in native cloud deployments of Cisco ISE

Cisco ISE instances deployed natively on cloud platforms do not support the upgrade workflow. Only new installations are supported. You can back up and restore configuration data. Cloud platforms that allow native deployment of Cisco ISE include:

1. Amazon Web Services (AWS)
2. Microsoft Azure Cloud
3. Oracle Cloud Infrastructure (OCI)

To upgrade the release on AWS from Cisco ISE release 3.2 to Cisco ISE release 3.3 :

1. Back up the configuration data from the Cisco ISE release 3.2 AWS instance.
2. Reconfigure the AWS instance with Cisco ISE release 3.3.
3. Restore configuration data on the newly created Cisco ISE release 3.3 instance.

Root CA chain regeneration

If any of these events occur, you must regenerate the root CA chain:

- Change the domain name or hostname of your Primary Administration Node (PAN) or PSN.
- Restore a backup on a new deployment.
- Promote the old primary PAN to a new primary PAN after an upgrade.

Regeneration process

Follow these steps to regenerate the root CA chain:

1. In the Cisco ISE GUI, navigate to **Administration** > **System** > **Certificates** > **Certificate Management** > **Certificate Signing Request**.
2. Click **Generate Certificate Signing Request (CSR)**.
3. From the **Certificate(s) will be used for** drop-down list, choose **ISE Root CA**.
4. Click **Replace ISE root CA Certificate Chain**.

Upgrade paths to Cisco ISE release 3.3

The upgrade process involves these high-level steps:

1. **Choose Upgrade Method:** Upgrade Cisco ISE using the GUI (recommended) method.
2. **Prepare System:** Verify prerequisites, backup configurations, and ensure resource availability.
3. **Perform Upgrade:** Upgrade nodes in the prescribed sequence.
4. **Post-Upgrade Tasks:** Validate system health and restore operational functionality.

When upgrading to Cisco ISE release 3.3, there are two possible approaches, depending on your current version:

- Single-step upgrade
- Two-step upgrade

Single-step upgrade

You can directly upgrade to Cisco ISE release 3.3 if your current Cisco ISE deployment is running on any of these releases:

- Cisco ISE release 3.0
- Cisco ISE release 3.1
- Cisco ISE release 3.2

You can upgrade in a single step without any intermediate upgrades.

Two-step upgrade

If your Cisco ISE deployment is running on a version **earlier than 3.0**, you must follow a two-step process:.

1. Upgrade your Cisco ISE to one of the supported intermediate Cisco ISE releases: 3.0, 3.1 or 3.2.
2. Once you are on one of these intermediate releases, you can then perform a single-step upgrade to Cisco ISE release 3.3.

For example, if you want to upgrade to Cisco ISE release 3.3 and you are currently on Cisco ISE release 3.1, you can upgrade directly to Cisco ISE release 3.3. However, if you are on Cisco ISE release 2.6, you must first upgrade to an intermediate version such as Cisco ISE release 3.0, 3.1, or 3.2, and then proceed to upgrade to Cisco ISE release 3.3.

Supported OS for VM

Cisco ISE runs on the Cisco Application Deployment Engine Operating System (ADE-OS), which is based on Red Hat Enterprise Linux (RHEL). For Cisco ISE release 3.3, ADE-OS is based on RHEL 8.4.

This table shows the RHEL versions used in different versions of Cisco ISE.

Table 1: RHEL versions for different Cisco ISE releases

Cisco ISE release	RHEL version
Cisco ISE release 3.1	RHEL 8.2
Cisco ISE release 3.2	RHEL 8.4
Cisco ISE release 3.3	RHEL 8.4

Note RHEL 8.2 and later support these VMware ESXi versions:

- VMware ESXi 6.7
- VMware ESXi 6.7 U1
- VMware ESXi 6.7 U2
- VMware ESXi 6.7 U3
- VMware ESXi 7.0
- VMware ESXi 7.0 U1
- VMware ESXi 7.0 U2
- VMware ESXi 7.0 U3

In addition to those previously mentioned, RHEL 8.4 supports newer compatible VMware ESXi versions.

Cisco ISE release 3.3 is the last release to support VMware ESXi 6.7.

For Cisco ISE release 3.1 and later releases, we recommend that you update to VMware ESXi 7.0.3 or later releases.

In the case of vTPM devices, you must upgrade to VMware ESXi 7.0.3 or later releases.

After upgrading Cisco ISE nodes on VMware VMs, turn off the VM to change the Guest OS to the supported RHEL version, then turn on the VM again.

Note If you select **Guest OS RHEL 8** and **Firmware EFI**, ensure that the **Enable UEFI Secure Boot** option is disabled in the **VM Options** tab. This option is enabled by default for Guest OS RHEL 8 VM. Ensure that you disable the **Enable UEFI Secure Boot** option for the Cisco ISE VM.

Upgrading Cisco ISE with the RHEL OS may take longer than usual due to possible changes in the Oracle database version, which require installing a new Oracle package during the upgrade.

Licensing information

This section provides licensing information for Cisco ISE release 3.3.

For more information on activating licenses in the Cisco ISE GUI, see Licensing.

Virtual appliance licenses

Cisco ISE release 3.1 and later supports the Cisco ISE VM license. This license replaces the VM Small, VM Medium, and VM Large licenses from earlier releases. The new ISE VM license covers Cisco ISE VM nodes for both on-premises and cloud deployments.

For more information, see "Cisco ISE Licenses" in the chapter "Licensing" in the *Cisco ISE Administrator Guide, Release 3.3*.

Specific license reservation

Specific license reservation is a smart licensing method for managing licenses in situations where security requirements prevent a persistent connection between Cisco ISE and the Cisco Smart Software Manager (CSSM). Specific license reservation allows you to reserve licenses on a Cisco ISE node.

You can create a specific license reservation by defining the type and number of licenses you need to reserve. Then, activate the reservation on a Cisco ISE node. The Cisco ISE node, where you register and enable the reservation, tracks license usage and enforces license compliance.

For more information, see "Specific license reservation" in the chapter "Licensing" in the *Cisco ISE Administrator Guide, Release 3.3*.

CHAPTER 2

Prepare for Upgrade

Prepare for upgrade

Before you start the upgrade process, complete these tasks:

Health check

Run a health check on your Cisco ISE deployment before upgrading to identify and resolve critical issues that may cause downtime. For more information, see "Health Check" in the “Troubleshoot” chapter in the *Cisco ISE Administrator Guide, Release 3.3*.

Optimize upgrade duration

Follow these guidelines to address issues in your current deployment during the upgrade process. This will help reduce overall downtime.

- Upgrade to the latest patch in the existing version before starting the upgrade.

> **Note** If you are upgrading from earlier releases and have an SSM On-Prem server configured, you must disconnect the SSM On-Prem server before you begin the upgrade process.

- Test the upgrade in a staging environment to identify and address any issues without impacting the live system.
    - All the nodes in the Cisco ISE deployment should be at the same patch level in order to exchange data.

> **Note** If the nodes in your deployment are not running the same Cisco ISE version and patch level, you will see a warning message: Upgrade cannot begin. This means the upgrade is blocked. Make sure all nodes have the same version and patch installed before starting the upgrade.

- Follow these guidelines if you are upgrading from a Cisco ISE release earlier than Cisco ISE release 3.2 patch 3.
    - Depending on the number of PSNs in your deployment and the staff available, install the required final version of Cisco ISE, apply the latest patch, and prepare the system for use.
    - If you want to retain the MnT logs, complete the tasks for MnT nodes and join them to the new deployment as MnT nodes. However, if you do not need to retain the operational logs, you can skip the step by reimaging the MnT nodes.
    - Cisco ISE installation can be performed in parallel in a multi-node deployment without impacting the production environment. Installing ISE servers in parallel saves time, especially when you are using backup and restore from a previous release.
    - PSN can be added to the new deployment to download the existing policies during the registration process from the PAN.
- It is best practice to archive old logs instead of transferring them to new deployments. Restoring operational logs on the MnT nodes does not synchronize them with other nodes if you change MnT roles later.
- If you have two data centers with full distributed deployment, first upgrade the backup data center. Then test the use cases before upgrading the primary data center.

- Download and store the upgrade software in a local repository before the upgrade to speed up the process.
- If you are currently upgrading to a recent Cisco ISE release, you can either use Health Check or Upgrade Readiness Tool (URT) to run system diagnosis before you initiate the upgrade process.
- Use the URT to detect and fix any configuration data upgrade issues before you start the upgrade process. Most upgrade failures occur due to configuration data issues during the upgrade. The URT validates the data to identify issues before the upgrade and fixes or reports them whenever possible. The URT is available as a separate downloadable bundle that can be run on a Secondary PAN or standalone node. There is no downtime to run this tool. This video explains how to use the URT: https://video.cisco.com/detail/video/5797832452001.

> **Warning** Do not run the URT on the Primary PAN. The URT tool does not simulate MnT operational data upgrades.

- When you upgrade Cisco ISE using the GUI, it shows a system generated timestamp on each node. If the URT takes additional time to complete the upgrade, we recommend using the CLI instead.
- Back up the load balancers before changing the configuration. You can remove the PSNs from the load balancers during the upgrade window and add them back after the upgrade.

- Disable automatic PAN failover (if configured) and disable the heartbeat between PANs during the upgrade.
- Review the existing policies and eliminate any outdated or redundant rules.
- Remove unwanted monitoring logs and endpoint data.
- You can take a backup of configuration and operations logs and restore it on a temporary server that is not connected to the network. You can use a remote logging target during the upgrade window.

  You can use these options after the upgrade to reduce the number of logs that are sent to MnT nodes and improve the performance:

  - Use the MnT collection filters to filter incoming logs and avoid duplication of entries in AAA logs. To view this window, navigate to **Administration** > **System** > **Logging** > **Collection Filters** in the Cisco ISE GUI.
  - You can create remote logging targets by navigating to **Administration** > **System** > **Logging** > **Remote Logging Targets** in the Cisco ISE GUI. You can then route each logging category to a specific logging target by navigating to **System** > **Logging** > **Logging categories** in the Cisco ISE GUI.
  - Enable the **Ignore Repeated Updates** options to avoid repeated accounting updates. To view this window, navigate to **Administration > System > Settings > Protocols > RADIUS** in the Cisco ISE GUI.

- Download and use the latest upgrade bundle for upgrade. Use this query in the Bug Search Tool to find the upgrade-related defects that are open and fixed: http://cs.co/ise-upgrade-bugsearch.
- Test all the use cases for the new deployment with fewer users to ensure service continuity.

Validate data

Cisco ISE includes URT. You can run this tool to detect and resolve data upgrade issues before starting the upgrade process.

Most upgrade failures occur because of data upgrade issues. Use URT to validate your data before an upgrade, identify and report issues, and fix issues when possible.

The URT is available as a separate downloadable bundle. Run the URT on a Secondary Administration Node (SAN) to provide high availability and support deployments with multiple nodes, or run it on the standalone node for a single-node deployment.

Warning In multiple-node deployments, do not run the URT on the Primary Policy Administration Node.

You can run the URT from the CLI of the Cisco ISE node. The URT

1. checks whether it is run on a supported version of Cisco ISE (supported versions are Cisco ISE release 3.0, 3.1, and 3.2),
2. verifies that URT is run on either a standalone Cisco ISE node or a Secondary Policy Administration Node (secondary PAN),
3. checks if the URT bundle is less than 45 days old to ensure you use the most recent URT bundle, and

4. checks all these prerequisites are met:
    - Version compatibility
    - Persona checks
    - Disk space

Note Verify the available disk size with Disk Requirement Size (see "Cisco Secured Network Server Series Appliances and Virtual Machine Requirements" chapter in the *Cisco ISE Installation Guide, Release 3.3*. If you need to increase the disk size, reinstall Cisco ISE and restore a configuration backup.

    - NTP server
    - Memory
    - System and trusted certificate validation

5. clones the configuration database,
6. copies the latest upgrade files to the upgrade bundle, and

Note If there are no patches in the URT bundle, the output returns `N/A`. This is expected behavior during the installation of a hot patch.

7. performs a schema and data upgrade on the cloned database.

    If the upgrade on the cloned database is successful, it provides an estimate of the time required for the upgrade to complete.

    If the upgrade is successful, the tool removes the cloned database.

    If the upgrade on the cloned database fails, the tool collects the required logs, prompts for an encryption password, generates a log bundle, and stores the bundle on the local disk.

Download and run the URT

The URT checks the configuration data before the upgrade to identify issues that could cause an upgrade failure.

Before you begin

While running the URT, do not simultaneously :

- Back up or restore data
- Make any persona changes

Procedure

Step 1 Create a repository, on page 11

Step 2 Install the URT, on page 11

Create a repository

Create a repository, then copy the URT bundle. For more information about creating a repository, see “Create repositories” in the chapter “Maintain and Monitor” in the *Cisco ISE Administrator Guide, Release 3.3*.

To improve performance and reliability, use FTP. Avoid using repositories located across slow WAN links. Choose a local repository near your nodes.

Before you begin

Make sure your connection to the repository has enough bandwidth.

Procedure

Step 1 Download the URT bundle from the Cisco ISE Download Software Center (ise-urtbundle-3.3.xxx-1.0.0.SPA.x86_64.tar.gz).

Step 2 Optionally, to save time, copy the URT bundle to the local disk on the Cisco ISE node using the command:

```
copy repository_url/path/ise-urtbundle-3.3.xxx-1.0.0.SPA.x86_64.tar.gz disk:/
```

To copy the upgrade bundle using SFTP, perform these steps:

```
(Add the host key if it does not exist) crypto host_key add host mySftpserver
copy sftp://aaa.bbb.ccc.ddd/ ise-urtbundle-3.3.xxx-1.0.0.SPA.x86_64.tar.gz disk:/
```

The value "aaa.bbb.ccc.ddd" represents the IP address or hostname of the SFTP server; "ise-urtbundle-3.3.*xxx*-1.0.0.SPA.x86_64.tar.gz" is the name of the URT bundle.

Install the URT

The URT identifies data issues that might cause an upgrade failure. It reports or fixes these issues where possible. To install the URT:

Before you begin

Storing the URT bundle on the local disk allows the installation process to complete more quickly.

Procedure

Step 1 Enter this command to install the tool:

```
application install ise-urtbundle-3.3.0.x.SPA.x86_64.tar.gz reponame
```

Note
If the application is not installed successfully, URT provides the reason for the upgrade failure. Fix any reported issues, then install the URT again.

Step 2 Remove the 5G attribute if you are upgrading to Cisco ISE release 3.3. For more information, see the "Configure Cisco Private 5G as a Service" section in the "Secure Access" chapter in the *Cisco ISE Administrator Guide, Release 3.3*.

Note
If you do not remove the 5G attribute, you will see this error:

```
Error Occurred while adding 5G field to access service

Error while applying changes in version: 3.3.0.100 class: com.cisco.cpm.acs.nsf.im.NetworkAccessUpgrade

com.cisco.cpm.infrastructure.upgrade.api.UpgradeFailureException:
com.cisco.cpm.nsf.api.exceptions.NSFEntityAttributeException: AccessService FIVEG
DuplicateAttributeException~AttributeName : FIVEG already exists as FIVEG

        at
com.cisco.cpm.acs.nsf.im.NetworkAccessUpgrade.upgradeAllowFiveG(NetworkAccessUpgrade.java:2671)

        at com.cisco.cpm.acs.nsf.im.NetworkAccessUpgrade.upgrade(NetworkAccessUpgrade.java:585)

        at
com.cisco.cpm.infrastructure.upgrade.impl.UpgradeServiceRegistrar.UpgradeServices(UpgradeServiceRegistrar.java:132)


        at
com.cisco.cpm.infrastructure.upgrade.impl.UpgradeServiceRegistrar.main(UpgradeServiceRegistrar.java:185)

Caused by: com.cisco.cpm.nsf.api.exceptions.NSFEntityAttributeException: AccessService FIVEG
DuplicateAttributeException~AttributeName : FIVEG already exists as FIVEG
```

Remove the URT

Uninstall URT if the installation failed or if the installed version is outdated. Follow these steps to uninstall URT.

Before you begin

Make sure URT appears in the installed programs list.

Procedure

Step 1 Enter this command in the CLI:

```
application remove urt
```

Step 2 At the prompt, enter **Y**:

```
Continue with application removal? (y/n)
```

This message is displayed when the uninstallation is complete:

```
Application successfully uninstalled
```

What to do next

After the uninstallation is successful, you can install URT, if required.

Rename authorization simple condition to avoid compound condition name conflict

Cisco ISE comes with several predefined authorization compound conditions. The upgrade process fails if your old deployment contains an authorization simple condition with the same name as a predefined authorization compound condition. Rename your authorization simple condition before the upgrade to prevent a name conflict with a predefined compound condition.

Here is a list of predefined authorization compound condition names that must not conflict with any authorization simple condition names during an upgrade to prevent upgrade failures:

- Compliance_Unknown_Devices
- Non_Compliant_Devices
- Compliant_Devices
- Non_Cisco_Profiled_Phones
- Switch_Local_Web_Authentication
- Catalyst_Switch_Local_Web_Authentication
- Wireless_Access
- BYOD_is_Registered
- EAP-MSCHAPv2
- EAP-TLS
- Guest_Flow
- MAC_in_SAN
- Network_Access_Authentication_Passed

Update VMware settings

If you are upgrading Cisco ISE nodes on VMs,

1. change the guest OS to RHEL 7,
2. power down the VM,
3. change the guest OS to RHEL 7, and
4. power on the VM.

RHEL 7 supports only E1000 and VMXNET3 network adapters. Change the network adapter type before you upgrade.

Update the sponsor group names

Before upgrading, rename any sponsor groups with non-ASCII characters to use only ASCII characters.

Cisco ISE does not support non-ASCII characters in sponsor group names.

Enable key firewall ports

If you have a firewall deployed between your PAN and any other node, you must open these ports before upgrading:

- TCP 1521: For communication between the PAN and monitoring nodes.
- TCP 443: For communication between the PAN and all other secondary nodes.
- TCP 12001: For global cluster replication.
- TCP 7800 and 7802: Required for PSN group clustering when PSNs are part of a node group.

For a full list of ports that Cisco ISE uses, see the chapter "Cisco ISE Ports Reference" in the *Cisco ISE Installation Guide, Release 3.3*.

Back up Cisco ISE configuration and operational data from the PAN

Obtain a backup of the Cisco ISE configuration and operational data from either the CLI or the GUI. To back up the configuration and operational data using the CLI, enter this command:

backup *backup-name* **repository** *repository-name* {**ise-config | ise-operational**} **encryption-key** {**hash | plain**} *encryption-keyname*

Note When Cisco ISE runs on VMware, VMware snapshots are not supported for backing up Cisco ISE data.

A VMware snapshot saves the status of a VM at a specific point in time. In a multi-node Cisco ISE deployment, all nodes continuously synchronize data with the current database. Restoring a snapshot might cause database replication and synchronization issues. Use the Cisco ISE backup functionality for data archival and restoration.

If you use VMware snapshots to back up Cisco ISE data, Cisco ISE services stop. To restore the ISE node, reboot it.

You can also obtain the configuration and operational data backup from the Cisco ISE Admin Portal. Ensure that you

- have created repositories that store the backup file
- do not use a local repository to back up data, and

  You cannot back up the monitoring data in the local repository of a Remote Monitoring node.
- do not use CD-ROM, HTTP, HTTPS, or TFTP repositories, because they are read-only or do not support file listing.

Perform these steps to obtain the back up from Cisco ISE GUI:

1. In the Cisco ISE GUI, navigate to **Administration** > **Maintenance** > **Backup and Restore**.
2. Click **Backup Now**.
3. Enter the values as required to perform a backup.
4. Click **OK**.
5. Verify that the backup completed successfully.

   Wait for the backup to finish before changing or promoting node roles. Changing node roles during backup shuts down all processes and may cause data inconsistencies.

After backup, verify the backup file exists in the specified repository. Cisco ISE appends the backup filename with a timestamp and adds a CFG tag for configuration backups and an OPS tag for operational backups.

Note Cisco ISE allows you to obtain a backup from a Cisco ISE node (A) and restore it to another Cisco ISE node (B), both having the same hostnames (but different IP addresses). However, after you restore the backup on node B, do not change the hostname of node B because it might cause issues with certificates and portal group tags.

Back up system logs from the PAN

Obtain a backup of the system logs from the PAN using the CLI. Use this CLI command:

backup-logs *backup-name* **repository** *repository-name* **encryption-key** { **hash | plain**} *encryption-key name*

CA certificate chain

Before upgrading to Cisco ISE release 3.3, ensure that the internal CA certificate chain is valid. Follow these steps to see if the internal CA certificate is valid:

1. In the Cisco ISE GUI, navigate to **Administration** > **System** > **Certificates** > **Certificate Authority** > **Certificate Authority Certificates**.
2. For each node in the deployment, select the certificate labeled **Certificate Services Endpoint Sub CA** in the **Friendly Name** field.
3. Click **View**.
4. Check if the `Certificate Status is good` message is visible.
5. Fix any broken certificate chains before upgrading Cisco ISE by navigating to **Administration** > **System** > **Certificates** > **Certificate Management** > **Certificate Signing Requests** > **ISE Root CA** in the Cisco ISE GUI.

Check certificate validity

If any certificate in the Cisco ISE Trusted Certificates or System Certificates store has expired, the upgrade fails.

Before you upgrade, follow these steps to check the validity of trusted certificates.

1. In the Cisco ISE GUI, navigate to **Administration** > **System** > **Certificates** > **Certificate Management** > **Trusted Certificates**.
2. Check the **Expiration Date** field of the certificate. Renew the certificate if its validity has expired or is about to expire.

Similarly, follow these steps to check the validity of system certificates.

1. In the Cisco ISE GUI, navigate to **Administration** > **System** > **Certificates** > **Certificate Authority** > **Certificate Authority Certificates**.
2. Check the **Expiration Date** field of the certificate. Renew the certificate if its validity has expired or is about to expire.

Delete a certificate

To delete an expired certificate, complete these steps:

Procedure

Step 1 In the Cisco ISE GUI, navigate to **Administration** > **System** > **Certificates** > **Certificate Management** > **System Certificates**.

Step 2 Identify and select the certificate that has expired.

Step 3 Click **Delete** to remove the selected certificate.

Step 4 In the Cisco ISE GUI, navigate to **Administration > System > Certificates > Certificate Management > Trusted Certificates**.

Step 5 Select the expired certificate.

Step 6 Click **Delete**.

Step 7 In the Cisco ISE GUI, navigate to **Administration > System > Certificates > Certificate Authority > Certificate Authority Certificates**.

Step 8 Select the expired certificate.

Step 9 Click **Delete**.

Export certificates from all nodes

Export all local certificates and their private keys from every node in your deployment to a secure location. Record the configuration for each certificate, including the service with which it is used.

Procedure

Step 1 In the Cisco ISE GUI, navigate to **Administration** > **System** > **Certificates** > **Certificate Management** > **System Certificates**.

Step 2 Select the certificate and click **Export**.

Step 3 Select the **Export Certificate and Private Key** radio button.

Step 4 Set a **Private Key Password** and then **Confirm Password**.

Step 5 Click **Export**.

The local certificates and their private keys are downloaded to your system.

Export certificates from Trusted Certificates Store

Export all certificates from the Trusted Certificates Store of the PAN. Record the configuration for each certificate, including the service with which the certificate is used.

Procedure

Step 1 In the Cisco ISE GUI, navigate to **Administration** > **System** > **Certificates** > **Certificate Management** > **Trusted Certificates**.

Step 2 Select the certificate and click **Export**.

Disable automatic failover and scheduled backups for upgrade

You cannot perform deployment changes when running a backup in Cisco ISE. Disable automatic configurations to ensure your upgrade goes smoothly. You should disable these configurations before you upgrade Cisco ISE:

- PAN Automatic Failover: Disable this option in the PAN before upgrading Cisco ISE.
- Scheduled backups: Disable all backup schedules before upgrading. After the upgrade, reschedule and recreate the backup schedules.

  Backups scheduled to run **once** are triggered every time the Cisco ISE application restarts. If you have a backup schedule set to run only once, disable it before upgrading.

Configure NTP server and verify availability

During upgrade, the Cisco ISE nodes reboot, migrate, and replicate data from the PAN to the SAN. For these operations, it is important that the NTP server in your network is configured correctly and is reachable. If the NTP server is not set up correctly or is unreachable, the upgrade process fails.

Ensure that the NTP servers in your network are reachable, responsive, and synchronized throughout the upgrade process.

Earlier versions of Cisco ISE use chrony instead of the Network Time Protocol daemon (ntpd). Ntpd synchronizes with servers that have a root dispersion up to 10 seconds, whereas chrony synchronizes with servers that have a root dispersion of less than 3 seconds. Therefore, we recommend that you use an NTP server with low root dispersion before upgrading to required Cisco ISE version to avoid NTP service disruption.

Upgrade your VM

Cisco ISE software must synchronize with both the chip and appliance capacity so that it can support the latest CPU and memory resources in UCS hardware.

As Cisco ISE versions progress,

- support for older hardware is phased out, and
- newer hardware is introduced.

It is recommended to upgrade your VM capacity to improve performance.

When planning VM upgrades, use OVA files to install the software efficiently. Each OVA file is a package that describes the VM and reserves the hardware resources required to install Cisco ISE software on your appliance.

For more information about the VM and hardware requirements, see "Hardware and Virtual Appliance Requirements for Cisco ISE" in the "Cisco Secured Network Server Series Appliances and Virtual Machine Requirements" chapter in the *Cisco ISE Installation Guide, Release 3.3*.

Cisco ISE VMs require dedicated resources in the VM infrastructure. These include an adequate number of CPU cores, comparable to hardware appliances, to ensure performance and scalability.

Resource sharing in the VM infrastructure can negatively impact performance. It may cause high CPU usage, delays in user authentication and registration, dropped logs, slow reporting, and reduced dashboard responsiveness.

Note Use reserved resources for CPU, memory, and hard disk space during upgrades instead of shared resources to avoid performance issues.

The local disk allocation has increased to 29 GB for newer versions of Cisco ISE. Therefore, Cisco ISE requires a minimum disk size of 300 GB for each VM.

Record profiler configuration

If you use the Profiler service, record the profiler configuration for each PSN from the Admin portal.

Follow these steps to find the profiler configuration information:

1. In the Cisco ISE GUI, navigate to **Administration** > **System** > **Deployment**.
2. Select the node.
3. On the **Edit Node** page, go to the **Profiling Configuration** tab.

4. Note the configuration information or capture screenshots.

Obtain Active Directory and internal administrator account credentials

If you use Active Directory (AD) as your external identity source, make sure you have your AD credentials and a valid internal administrator account ready. After the upgrade, your AD connection might be lost. If that happens, use your Cisco ISE internal administrator account to log in to the Admin portal and use your Active Directory credentials to rejoin Cisco ISE to AD.

Activate Mobile Device Management vendor

If your deployment works with third-party Mobile Device Management (MDM) solutions, ensure that the MDM vendor status is active before upgrading.

If an MDM server name is used in an authorization policy and the corresponding MDM server is disabled, the upgrade process fails. As a workaround, you can do one of these:

1. Enable the MDM server before upgrading.
2. Delete the condition that uses the MDM server name attribute from the authorization policy.

Create repository and copy the upgrade bundle

Create a repository to obtain backups and copy the upgrade bundle. For more information, see "Create Repositories" in the "Maintain and Monitor" chapter in the *Cisco ISE Administrator Guide, Release 3.3*.

Perform these steps to create a repository and copy the upgrade bundle:

1. Place the upgrade bundle on the local disk to save time during the upgrade process. You can also use this command to copy and extract the upgrade bundle on the local disk:

   ```
   application upgrade prepare <upgrade bundle name> <repository name>
   ```

Note

- Ensure your connection to the repository is fast and stable. If the upgrade bundle (about 14 GB) takes more than 35 minutes to download to the node, the process will time out.
- If you store configuration files on a local disk, these files are deleted during the upgrade. Create a Cisco ISE repository and copy them to the repository to keep your files.
- Choose a local repository near your nodes instead of one on a slow WAN link. Use FTP for faster performance and reliability.

2. Download the upgrade bundle from Cisco.com.

   To upgrade to Cisco ISE release 3.3, use this upgrade bundle: ise-upgradebundle-3.0.x-3.2.x-to-3.3.0.*xxx*.SPA.x86_64.tar.gz

3. To perform the upgrade, copy the upgrade bundle to the Cisco ISE node local disk using this command:

   copy *repository_url/path/ise-upgradebundle-3.0.x-3.2.x-to-3.3.0.xxx.SPA.x86_64.tar.gz* **disk:/**

For example, if you want to use SFTP to copy the upgrade bundle, you can do this:

1. (Add the host key if it does not exist) **crypto host_key add host** *mySftpserver*
2. **copy sftp://***aaa.bbb.ccc.ddd/ise-upgradebundle-3.0.x-3.2.x-to-3.3.0.xxx.SPA.x86_64.tar.gz* **disk:/**

   aaa.bbb.ccc.ddd is the IP address or hostname of the SFTP server and *ise-upgradebundle-3.0.x-3.2.x-to-3.3.0.xxx.SPA.x86_64.tar.gz* is the name of the upgrade bundle.

Check the available disk space

Ensure that you allocate the required disk space for VMs. For more information, see "Disk space requirements for VMs in a Cisco ISE deployment" in Cisco Secured Network Server Series Appliances and Virtual Machine Requirements chapter in the *Cisco ISE Installation Guide, Release 3.3*.

If you need to increase the disk size, reinstall Cisco ISE and restore a configuration backup.

Check load balancer configuration

If you use a load balancer between the PAN and the PSN, set the session timeout on the load balancer high enough so it does not disrupt the upgrade process. A session timeout that is too short can interrupt the upgrade on PSNs. For instance, if a session ends during a database transfer from the PAN to a PSN, the upgrade on the PSN fails.

Resizing MnT hard disk

Upgrading does not require changes to the MnT disk capacity. If your logs consistently reach capacity and you need additional hardware resources, plan the MnT hard disk size based on your log retention needs. Log retention capacity has increased significantly over recent Cisco ISE releases.

You can activate collection filters to remove unnecessary logs from different devices. Excess logs may overwhelm your Cisco ISE MnT.

In the Cisco ISE GUI, navigate to **Administration** > **System** > **Logging** > **Collection filters**.

Refer to the Cisco ISE storage requirements on the Cisco ISE Performance and Scalability community page. The table provides log retention information that is based on the number of endpoints for RADIUS and the number of network devices for TACACS+. Calculate log retention separately for TACACS+ and RADIUS.

CHAPTER 3

Perform the Upgrade

Upgrade sequence of the nodes

You can upgrade Cisco ISE using the GUI (recommended), the backup and restore feature, or the CLI.

If you use the GUI (recommended) method to upgrade, you can select the order in which nodes are upgraded. Upgrade the nodes in this order to minimize downtime, maximize resiliency, and make rolling back easier.

Complete these tasks before starting the upgrade:

- Back up all configuration and monitoring data.
- Export the internal CA key and certificate chain.
- Back up server certificates for all Cisco ISE nodes.

The upgrade process for nodes occurs in this order:

1. SAN

   At this point, the PAN remains at the previous version and can be used for rollback if the upgrade fails.

2. Primary Monitoring Node or Secondary Monitoring Node

   If you have a distributed deployment, upgrade the nodes available in the site with the SAN.

3. PSNs

   If you are upgrading from an earlier Cisco ISE release to a recent release using the GUI, you can select a group of PSNs to be upgraded simultaneously. This will reduce the overall upgrade downtime.

   After upgrading a set of PSNs, verify the success of the upgrade (see Verify the upgrade process, on page 45) and run network tests to ensure the new deployment works as expected. If the upgrade is successful, you can upgrade the next set of PSNs.

4. Secondary Monitoring Node or Primary Monitoring Node

5. PAN

After upgrading the PAN, rerun upgrade verification and network tests.

Note If the upgrade fails when registering the PAN, the system initiates a rollback and changes the node to standalone mode. Use the CLI to upgrade the node as a standalone. Then register it to the new deployment as a SAN.

After the upgrade, the SAN becomes the PAN, and the original PAN becomes the SAN. In the **Edit Node** window, click **Promote to Primary** to make the SAN the PAN, if needed.

If the administration nodes also have the monitoring persona, use the node sequence shown in this table.

Table 2: Node personas and their upgrade sequence

Node personas in the current deployment	Upgrade sequence
SAN/Primary Monitoring Node, PSN, PAN/Secondary Monitoring Node	**1.** SAN/Primary Monitoring Node **2.** PSN **3.** PAN/Secondary Monitoring Node
SAN/Secondary Monitoring Node, PSN, PAN/Primary Monitoring Node	**1.** SAN/Secondary Monitoring Node **2.** PSN **3.** PAN/Primary Monitoring Node
SAN, Primary Monitoring Node, PSN, PAN/Secondary Monitoring Node	**1.** SAN **2.** Primary Monitoring Node **3.** PSN **4.** PAN/Secondary Monitoring Node
SAN, Secondary Monitoring Node, PSN, PAN/Primary Monitoring Node	**1.** SAN **2.** Secondary Monitoring Node **3.** PSN **4.** PAN/Primary Monitoring Node
SAN/Primary Monitoring Node, PSN, Secondary Monitoring Node, PAN	**1.** SAN/Primary Monitoring Node **2.** PSN **3.** Secondary Monitoring Node **4.** PAN

Node personas in the current deployment	Upgrade sequence
SAN/Secondary Monitoring Node, PSNs, Primary Monitoring Node, PAN	**1.** SAN/Secondary Monitoring Node **2.** PSN **3.** Primary Monitoring Node **4.** PAN

You will get an error message "No SAN in the Deployment" under these circumstances:

- There is no SAN in the deployment.
- The SAN is down.
- The SAN is upgraded and moved to the upgraded deployment. This occurs when you use the **Refresh Deployment Details** option after upgrading the SAN.

To resolve this issue, complete one of these tasks:

- If the deployment does not have a SAN, configure a SAN and retry upgrade.
- If the SAN is down, bring up the node and retry the upgrade.
- If the SAN is upgraded and moved to the upgraded deployment, use the CLI to manually upgrade the other nodes in the deployment.

Choose your upgrade method

You can choose an upgrade process based on your technical expertise and the time available for the upgrade. This release of Cisco ISE supports these upgrade processes:

- Upgrade using the GUI (recommended)
- Upgrade using backup and restore (limited to Cisco ISE release 3.2 patch 2)
- Upgrade using the CLI

This table compares Cisco ISE upgrade methods.

Table 3: Cisco ISE upgrade method comparison

Comparison factors	Upgrade using the GUI (recommended)	Upgrade using backup and restore (limited to Cisco ISE release 3.2 patch 2)	Upgrade using the CLI
Process Type	Long	Fast	Longer
Administration required	Less	More	More
Difficulty level	Easy	Hard	Moderate

Comparison factors	Upgrade using the GUI (recommended)	Upgrade using backup and restore (limited to Cisco ISE release 3.2 patch 2)	Upgrade using the CLI
VMs	Each PSN is upgraded in parallel.	If there is enough capacity, new VMs can be prestaged and joined immediately to the new PAN.	Each PSN is upgraded, however, they can be done in parallel.
Upgrade time	Less (because each PSN is upgraded in parallel).	Least (because PSNs are imaged with new version instead of being upgraded).	Less (because each PSN is upgraded in parallel)
Personnel required	Fewer manual interventions are required because the upgrade process is automated.	Stakeholders from multiple business units transfer configuration settings and operational logs.	Technical expertise on Cisco ISE is required.
Rollback options	Easy	Difficult (requires reimaging of the nodes)	Easy

Upgrade using the GUI (recommended)

Overview

Starting with Cisco ISE release 3.2 patch 3, upgrading through the GUI is the recommended method.

You can also upgrade Cisco ISE from the GUI in a single click with some customizable options. Follow these steps to perform the upgrade:

1. In the Cisco ISE GUI, navigate to **Administration** > **System** > **Upgrade & Rollback**.
2. Create a new repository to download the upgrade bundle.

Note This method is not supported on cloud platforms such as Oracle Cloud Infrastructure (OCI), Amazon Web Services (AWS), and Azure Cloud Services.

Why upgrade using the GUI (recommended)

This method is recommended if you are upgrading from Cisco ISE release 3.2 patch 3 or later.

Here is why you should upgrade using the GUI (recommended) method:

- During the upgrade, the Secondary PAN is moved into an upgraded deployment automatically and is upgraded first, followed by Primary MnT. As a result, if either of these upgrades fail, it is mandatory that the node will be rolled back to the previous version and rejoin to the previous Cisco ISE deployment. You can select multiple PSNs in a batch and upgrade them simultaneously.
- In case of an upgrade failure, you can also choose to continue or cease the upgrade. This will result in a dual version of same Cisco ISE deployment, allowing for troubleshooting to occur before the upgrade

continues. Once all PSNs are upgraded, the Secondary MnT and Primary PAN is upgraded and joined to the new Cisco ISE deployment.

- Because this upgrade process requires limited technical expertise, a single administrator can start the upgrade. You can then assign NOC or SOC engineers to monitor and report the upgrade status or open a TAC case.

Advantages

Upgrading from Cisco ISE release 3.2 patch 3 or a later version using the GUI method offers these advantages:

- The upgrade is automated with minimal intervention.
- You can choose the upgrade order of the PSNs to ensure continuity whenever possible, especially when redundancy is available between data centers.
- A single administrator can execute the upgrade without assistance from additional personnel, third-party hypervisors, or network access devices.

Key considerations before the upgrade

You should consider these points before upgrading from Cisco ISE release 3.2 patch 3 or a later version using the GUI (recommended) method.

- **Continuation in failure scenarios:** In case of an upgrade failure, you can also choose to continue or cease the upgrade. This will result in a dual version of same Cisco ISE deployment, allowing for troubleshooting to occur before the upgrade continues. While the Cisco URT should indicate any incompatibilities or misconfigurations, if the **Proceed** field is checked, additional errors may be encountered if due diligence was not acted upon before the upgrade.
- **Rollback mechanism:** If an upgrade fails on a PAN or MnT node, the nodes are automatically rolled back. However, if a PSN fails to upgrade, the nodes remain on the same Cisco ISE version and can be fixed while impairing redundancy. Cisco ISE is still operational during this time, and therefore rollback abilities are limited without reimaging.
- **Time required:** Each PSN takes around 90-120 minutes to upgrade. When you upgrade a large number of PSNs sequentially, the process takes a significant amount of time. Upgrading in batches reduces the overall time required.

Best practices for the upgrade process

Here is the list of best practices for upgrading from Cisco ISE release 3.2 patch 3 or a later version using the GUI (recommended) method:

- If you have a large number of PSNs, group them into batches. Upgrade each batch separately.
- Cisco ISE offers a GUI-based centralized upgrade from the Admin portal. The upgrade process is much simplified, and the progress of the upgrade and the status of the nodes are displayed on the screen.
- Begin the upgrade only when the nodes are in the active state.

  In the Cisco ISE GUI, navigate to **Administration** > **System** > **Upgrade** > **Overview**. This menu option lists all the nodes in your deployment, the personas that are enabled on them, the version of ISE installed, and the status (indicates whether a node is active or inactive) of the nodes.

GUI-based upgrade options

Depending on your deployed Cisco ISE, you can select one of these options in the **Administration** > **System** > **Upgrade** > **Upgrade Selection** page to upgrade your Cisco ISE deployment:

- Full upgrade
- Split upgrade

Note Consider these pointers when upgrading Cisco ISE using GUI-based upgrade options:

- In Cisco ISE release 3.2 and later, you must use both the URT and UI prechecks with the Split Upgrade method. For the Full Upgrade method, only UI prechecks are required.
- Although these GUI upgrade methods are available from earlier releases onwards, you must run at least Cisco ISE 2.7 patch 4 to upgrade to Cisco ISE 3.2 and later.
- Do not install or roll back a patch on any node using the CLI while another upgrade is in progress through the GUI or CLI upgrade options.

Full upgrade

Full upgrade is a multistep process that

- upgrades all Cisco ISE nodes simultaneously
- completes the upgrade faster than the split upgrade process
- makes the application services unavailable as all nodes are upgraded in parallel
- supported for all latest Cisco ISE releases.

To upgrade Cisco ISE using **Full upgrade** option, complete these steps:

1. Start the upgrade, on page 26
2. Initiate prechecks, on page 27
3. Start the staging process, on page 29

Start the upgrade

To start the upgrade for your Cisco ISE deployment, complete these steps:

Procedure

Step 1 In the Cisco ISE GUI, navigate to **Administration** > **System** > **Upgrade & Rollback**.

Step 2 In the **Upgrade Selection** window, click **Full Upgrade**.

Step 3 Click **Initialize**.

Step 4 Click **Next** in the **Welcome** window to start the upgrade workflow.

Step 5 To avoid any blockers or downtime during the upgrade process, complete all the tasks listed in the **Checklist** window.

Step 6 Click **Print Checklist** (optional) to download the checklist for your reference.

Step 7 Check the **I have reviewed the checklist** check box after you have verified the items listed in the upgrade checklist. Click **Next**.

The **Prepare to Upgrade** window appears.

Initiate prechecks

Follow these steps to initiate the prechecks for your Cisco ISE deployment:

Procedure

Step 1 From the **Repository** drop-down list, choose the repository where your upgrade bundle is stored.

Step 2 From the **Bundle** drop-down list, choose the upgrade bundle.

Step 3 All the patch releases are listed in the **Patch** drop-down list. Choose the patch for the Cisco ISE release to which you are upgrading.

Step 4 Click **Start Preparation** to validate all the Cisco ISE components and to generate a report for your deployment.

During the upgrade process, Cisco ISE performs several checks.

Precheck list	Description
Repository Validation	Checks whether the repository is configured for all nodes.
Bundle Download	Downloads and prepares the upgrade bundle for all nodes.
Memory Check	Checks whether 25 percent memory space is available on the PAN or standalone node, and one GB memory space is available in all the other nodes.
PAN Failover Validation	Checks if PAN high-availability is enabled. Before you start the upgrade, you will receive a notification that PAN high availability will be disabled.
Scheduled Backup Check	Checks whether the scheduled backup is enabled. **Note** This check is not mandatory for the upgrade process.
Config Backup Check	Checks whether the configuration backup was completed recently. You can only start the upgrade process after you complete the backup.
Configuration Data Upgrade	Upgrades the configuration data on the database clone and creates an upgraded data dump. This check starts after you download the bundle.

Precheck list	Description
Platform Support Check	Checks if your deployment uses supported platforms. The system verifies that hardware includes at least 12-core CPUs, a 300-GB hard disk, and 16-GB memory, and that you use ESXi version 6.5 or later.
Deployment Validation	Checks if the deployment node is in sync or still in progress.
DNS Resolvability	Verifies that both forward and reverse lookups for host name and IP address are working correctly.
Trust Store Certificate Validation	Checks if the trust store certificate is valid or expired.
System Certificate Validation	Checks if each node's system certificate is properly validated.
Disk Space Check	Checks if the hard disk has enough free space so you can continue the upgrade process.
NTP Reachability and Time Source Check	Checks if you configured NTP and verifies the time source is an NTP server.
Load Average Check	Checks the system load at defined intervals. You can set the frequency to one, five, or 15 minutes.
Services or Process Failures	Shows you whether the service or application is running or has failed.

If a component is inactive or has failed, the system displays it in red. You will receive troubleshooting suggestions. Depending on the criticality of the component failure, you can proceed with the upgrade process, or you must resolve the issue before continuing.

Field	Description
Refresh Failed Checks	This option refreshes only the failures highlighted in red. You must resolve these failures before performing an upgrade. Warnings highlighted in orange do not prevent the upgrade, but they may affect Cisco ISE functionality later. Click the **Refresh** icon next to each warning to update the checks after resolving issues.
Expand to Show	Click this icon to see additional information about each node and its status.
Information	Click this icon to see more information about each component.
Download Report	Click this option to get a copy of the generated reports.

Cisco ISE displays the estimated time required to stage and upgrade your deployment. The calculation uses these factors:

- Network speed
- Node configuration, including the number of processors, RAM, and hard disk.
- Database data size
- Time taken by the node to start the application server.

Note
All the prechecks, except the **Bundle Download** and **Configuration Data Upgrade** checks, expire automatically three hours after system validation is initiated.

Start the staging process

Before you begin

Follow these steps to start the staging process for your Cisco ISE deployment:

Procedure

Step 1 After you complete the prechecks for all nodes, click **Start Staging** to begin the staging process.

During upgrade staging, the system copies the upgraded database file to all nodes. The system also backs up configuration files for each node.

If upgrade staging on a node is successful, the system displays it in green. If upgrade staging fails for a node, the system displays it in red and provides troubleshooting suggestions.

Click the **Refresh Failed Nodes** icon to reinitiate the upgrade staging for the failed nodes.

Step 2 Click **Next** to proceed to the **Upgrade Nodes** window.

The **Upgrade Nodes** window shows the overall upgrade progress and the status for each node in your deployment.

Step 3 Click **Start** to initiate the upgrade process.

Just before the upgrade procedure completes, the system displays the message:

```
The system is about to upgrade. Logging out.
```

Step 4 Click **OK** to proceed.

Note
You can log in to the secondary PAN again to monitor the progress of the upgrade.

Use the secondary PAN dashboard to monitor the primary PAN upgrade status. After you upgrade the primary PAN, use the primary PAN dashboard to monitor all Cisco ISE nodes.

An active Cisco ISE Tier License is required to monitor the upgrade process. The Cisco ISE Evaluation License is sufficient to view the upgrade process. You can view and monitor the upgrade process only if an active license is present. For more information, see "Licensing" in the *Cisco ISE Administrator Guide, Release 3.3*.

Note
To view the **Summary** window later, do not click the **Exit Wizard** option in this window.

Step 5 Click **Next** in the **Upgrade Nodes** window to check whether all the nodes are upgraded successfully.

If any nodes fail, a dialog box displays information about the failed nodes.

Step 6 Click **OK** in the dialog box to deregister the failed nodes from the deployment.

After completing the upgrade process, view and download the diagnostic upgrade reports for your deployment in the **Summary** window. Verify and download the upgrade summary reports containing details such as

- **Checklist**
- **Prepare to Upgrade**
- **Upgrade Report**, and
- **System Health** checklist items.

Split upgrade from the GUI

Split upgrade is a multistep process that

- upgrades your Cisco ISE deployment while other services remain available to you
- updates the network or load balancers to keep nodes available for authentication
- can limit downtime by dividing nodes into batches and upgrading each batch in sequence
- might take longer than a full upgrade
- can be used for upgrading from Cisco ISE release 3.2 patch 3 and later to Cisco ISE release 3.3 and later.

A new split upgrade framework was introduced in Cisco ISE release 3.2 patch 3 and later releases to improve stability and reduce downtime.

The new split upgrade workflow provides these advantages:

- Prechecks are completed before the proceeding phase of the upgrade.
- The data upgrade occurs during the precheck phase. This process reduces the upgrade time and prevents the system from becoming unusable due to data upgrade issues.
- You can select PSNs and MnT nodes during the first iteration with the secondary PAN or during the last iteration with the primary PAN. The selected MnT nodes can be primary or secondary.
- If you select PSNs and MnT nodes in the same iteration, they are upgraded simultaneously, which further reduces upgrade time.

To upgrade Cisco ISE using the Split Upgrade, complete these steps:

1. Prepare for upgrade, on page 30
2. Initiate preparation, on page 31
3. Start staging, on page 33

Prepare for upgrade

Follow these steps to prepare for Split Upgrade:

Procedure

Step 1 In the Cisco ISE GUI, navigate to **Administration** > **System** > **Upgrade**.

Step 2 In the **Upgrade Selection** window, click **Split Upgrade**. A warning message appears. The message states that the Precheck Report will be cleared if the upgrade type is changed to Split Upgrade. It also states that ISE Services will continue to work during the Precheck Execution and Staging Nodes stages. Click **OK** to continue.

Step 3 Click **Start Upgrade**.

Step 4 Click **Let's Do It** in the **Welcome** window to start the upgrade workflow.

Step 5 Complete all the tasks listed in the **Checklist** window to avoid delay or downtime during the upgrade process.

Step 6 (Optional) Click **Print Checklist** to download the checklist for your reference.

Step 7 After you have verified the items listed in the upgrade checklist, check the **I have reviewed the checklist** check box. Then click **Next**.

The **Node Selection** window appears.

Step 8 Check the check boxes next to the nodes that you want to upgrade in the current iteration.

Note

- The secondary PAN is selected by default in the first iteration of the upgrade. The primary PAN cannot be selected during the first iteration or in subsequent iterations until all other nodes are upgraded or selected for upgrade. For the first iteration, you may select the primary MnT, the secondary MnT, and any number of PSNs.
- If nodes other than the secondary PAN are selected in the first iteration, the upgrade occurs in two batches. The secondary PAN is upgraded in the first batch. The remaining selected nodes are then upgraded simultaneously in the second batch.
- The PSNs and MnT nodes can be upgraded in parallel in any iteration.
- It is recommended that you select 15 nodes or fewer per iteration during the upgrade process.

Step 9 Click **Next.**

The **Prepare to Upgrade** window opens.

Initiate preparation

Follow these steps to initiate preparation for Split Upgrade:

Procedure

Step 1 From the **Repository** drop-down list, choose the repository in which your upgrade bundle is stored.

Note
If you use an SFTP repository, add the crypto host key on all nodes using the CLI.

Step 2 From the **Bundle** drop-down list, choose the upgrade bundle.

All the patch releases are listed in the **Patch** drop-down list. Choose the latest patch for the Cisco ISE release you are upgrading to.

Step 3 Click **Start Preparation** to validate all the Cisco ISE components and to generate a report for your deployment.

Cisco ISE checks these during the upgrade process.

Note

- The **Bundle Download** is triggered on all the selected nodes, including the primary PAN during the first iteration. During later iterations, it is triggered only for the selected nodes.
- The **Scheduled Backup Check** and **Config Backup Check** happens only in the first iteration.

Table 4: This table describes each precheck and its corresponding function:

Precheck list	Description
Repository Validation	Checks whether the repository is configured for all the nodes.
Bundle Download	Downloads and prepares the upgrade bundle for all the nodes.
Memory Check	Checks whether 25 percent of memory space is available on the PAN or standalone node, and 1 GB of memory space is available in all the other nodes.
Patch Bundle Download	Downloads the patch bundle for the selected nodes.
PAN Failover Validation	Checks whether PAN high availability is enabled. You receive a notification that PAN high availability will be disabled before the upgrade begins.
Scheduled Backup Check	Checks whether the scheduled backup is enabled. **Note** This check is not mandatory for the upgrade process.
Config Backup Check	Verify that you have completed a configuration backup recently. The upgrade runs only after you complete the backup.
Configuration Data Upgrade	Runs the configuration data upgrade on the configuration database clone and creates the upgraded data dump. This check starts after the bundle download.
Platform Support Check	Checks the supported platforms in the deployment. It checks whether the system has at least 12-core CPUs, a 300-GB hard disk, and 16-GB memory. It also checks whether the ESXi version is 6.5 or later.
Deployment Validation	Checks the state of each deployment node (whether it is in sync or in progress).
DNS Resolvability	Checks the forward and reverse lookup of host names and IP addresses.
Trust Store Certificate Validation	Checks whether the trust store certificate is valid or has expired.
System Certificate Validation	Checks the system certificate validation for each node.
Disk Space Check	Checks whether the hard disk has enough free space to continue with the upgrade process.

NTP Reachability and Time Source Check	Checks whether NTP is configured in the system and whether the time source is from the NTP server.
Load Average Check	Checks the system load at specified intervals: one minute, five minutes, or 15 minutes.
Services or Process Failures	Indicates the state of the service or application (whether it is running or in a failed state).

If any of the components are inactive or have failed, they are displayed in red. You will receive troubleshooting suggestions. If the failed component is critical to the upgrade, resolve the issue before continuing. Otherwise, you may proceed with the upgrade.

Table 5: This table describes each field and its corresponding function:

Field	Description
Refresh Failed Checks	This option refreshes only the failures highlighted in red. You must fix these failures before performing an upgrade. Warnings highlighted in orange do not stop the upgrade but may affect Cisco ISE functions after the upgrade. Click the **Refresh** icon next to each warning message to refresh these checks after resolving the issues.
Expand to Show	Click this icon to see additional information about each node and its status.
Information	Click this icon to view details about each component.
Download Report	Click this option to get a copy of the generated reports.

You can view the estimated time for the upgrade. The system determines the estimated upgrade time using

- Cisco ISE installation time
- Node hardware specifications: number of processors, RAM, and hard disk
- Data size in database, and
- Time taken by the node to start the application server.

Note
All the prechecks, except the **Bundle Download** and **Configuration Data Upgrade** checks, expire automatically after three hours of initiating the system validation.

Local prechecks are run on all nodes during the first iteration. However, in subsequent iterations, these checks are run only on the selected nodes.

Start staging

Follow these steps to start the staging process:

Procedure

Step 1 After completing the prechecks for all nodes, click **Start Staging** to begin the staging process.

The **Upgrade Staging** window opens.

During upgrade staging, the upgraded database file is copied to all nodes in the iteration. The system also backs up the configuration files on those nodes.

If upgrade staging on a node is successful, a green indicator appears. If upgrade staging fails for a particular node, a red indicator appears. Troubleshooting suggestions are provided as well.

Click the **Refresh Failed Nodes** icon to reinitiate the upgrade staging for the failed nodes.

Step 2 Click **Next** to proceed to the **Upgrade Nodes** window.

In the **Upgrade Nodes** window, you can see the overall upgrade progress and the status for each node in your deployment.

Step 3 Click **Start** to initiate the upgrade process.

The upgrade progress can be monitored from the secondary PAN GUI when the primary PAN is getting upgraded, and from the primary PAN GUI when the secondary PAN is getting upgraded.

Note
To monitor the upgrade process, you must have an active Cisco ISE Tier License. To view the upgrade process, the Cisco ISE Evaluation License is sufficient. If no active licenses are present, you cannot view or monitor the upgrade process. For more information on Cisco ISE licenses, see the chapter "Licensing" in the *Cisco ISE Administrator Guide* for your release.

If you click the **Exit Wizard** option in this window, you will not be able to view the **Summary** window later.

Step 4 Click **Next** in the **Upgrade Nodes** window to check whether all the nodes are upgraded successfully.

If there are any failed nodes, a dialog box with information about the failed nodes is displayed.

Step 5 Click **Finish** in the **Summary** window.

You are redirected to the **Node Selection** window, so that you can select the nodes for the next iteration.

Step 6 Continue with the next iteration by using the same sequence until all the nodes are upgraded.

The **Configuration Data Dump Generation** precheck is run instead of the **Configuration Data Upgrade** precheck in all the iterations of the process, except the first.

You do not need to upgrade all nodes in a deployment during the split upgrade process. You may stop after any number of iterations. However, you must clear the **Upgrade Tables** in the new deployment using the CLI commands **application configure ise**, then **reset upgrade tables** from the admin shell.

After the upgrade process is completed, you can view and download the diagnostic upgrade reports that are for your deployment in the **Summary** window. The upgrade summary reports include relevant details such as the old and new personas of the upgraded nodes, the prechecks performed, and the **Prepare to Upgrade** and **Prepare to Upgrade** checklist items.

If you have carried out the Cisco ISE split upgrade, the secondary PAN is promoted to the primary PAN in the process. In the Cisco ISE administration portal, choose **Administration** > **Licensing**. In the Cisco Smart Licensing area, click **Update**. A licensing alarm appears in your Cisco ISE until you update your licenses.

Upgrade using backup and restore (limited to Cisco ISE release 3.2 patch 2)

Overview

The backup and restore method is recommended for Cisco ISE releases up to and including version Cisco ISE release 3.2 patch 2.

Reimaging of the Cisco ISE node is performed during initial deployment and troubleshooting. For upgrades to Cisco ISE release 3.2 patch 2 or earlier, you may also reimage the Cisco ISE node to upgrade a deployment, which allows restoration of the policy onto the new deployment after the new version is installed.

If resources are limited and the new deployment cannot support a parallel Cisco ISE node, the Secondary PAN and MnT are removed from the production deployment and upgraded before other nodes. The nodes are then moved into the new deployment. A configuration and operational backup from the previous deployment is restored on the respective nodes, creating a parallel deployment. This process restores policy sets, custom profiles, network access devices, and endpoints to the new deployment without manual intervention.

Important The backup and restore upgrade method is supported for all Cisco ISE releases and patches. However, for Cisco ISE releases prior to and including 3.2 patch 2, the backup and restore method was recommended as the primary upgrade method. This method involves creating configuration and operational backups of the existing deployment and restoring them on the new deployment, which helps prevent data loss and reinstates node settings.

For Cisco ISE versions later than 3.2 patch 2, while the backup and restore method remains supported, the GUI-based upgrade method is recommended instead. The GUI-based upgrade offers enhanced features such as improved monitoring and a faster upgrade process compared to backup and restore.

In case of VM failure or issues during upgrade in later versions, recovery involves reimaging the nodes or using the rollback procedures supported by the GUI or CLI upgrade methods.

Why upgrade using backup and restore method

Cisco recommends the backup and restore method for releases up to and including Cisco ISE release 3.2 patch 2. This procedure starts by creating configuration and operational backups of the existing Cisco ISE deployment and then applying them to the new deployment.

This method helps reinstate your current Cisco ISE deployment node settings and prevents data loss if any issues occur during the upgrade.

Advantages

Upgrading Cisco ISE using backup and restore method (recommended upto Cisco ISE release 3.2 patch 2) offers these advantages:

- You can restore the configuration settings and the operational logs from the previous Cisco ISE deployment, which prevents data loss.
- You can select which nodes to reuse for the new deployment.
- You can upgrade multiple PSNs in parallel, reducing the upgrade downtime.
- You can stage the nodes outside maintenance windows, reducing the upgrade time during production.

Key considerations before the upgrade

Review these considerations before upgrading Cisco ISE using the backup and restore method (recommended upto Cisco ISE release 3.2 patch 2).

- Resources required: The backup and restore upgrade process requires additional resources. Reserve these resources for the Cisco ISE deployment before releasing them. If you reuse existing hardware, you must balance the additional load across the nodes that remain online. Therefore, evaluate the current load and latency limits before deployment. This evaluation ensures the deployment can support an increased number of users per node.
- Personnel required: You will need resources from multiple business units to perform the upgrade. These units include network administration, security administration, data center, and virtualization teams. In addition, you must rejoin the node to the new deployment, restore certificates, rejoin to Active Directory, and wait for policy synchronization. These actions can lead to multiple reloads and require a timeframe similar to that of a net-new deployment.
- Rollback mechanism: When you reimage the nodes, all information and configuration settings from the previous deployment are erased. Therefore, the rollback mechanism for a backup and restore upgrade follows the same procedure as reimaging the nodes again.

Best practices for the upgrade process

This section presents best practices for upgrading Cisco ISE using the backup and restore method (recommended upto Cisco ISE release 3.2 patch 2).

- Create a standalone environment or dedicate load balancers that switch the virtual IP address for RADIUS requests.
- You can start the deployment process well before the maintenance window. Then, point the user load balancer to the new deployment.
- If you use RSA SecurID Identity Sources, when you add a new PSN, you must generate a new configuration file with all the PSNs at the primary instance of your RSA Authentication Manager.

Note To avoid generating a new RSA configuration every time you add a new PSN, you must know the IP address of all the nodes that you are going to add to the deployment before starting the backup and restore process. Then, you must generate the RSA configuration file using all the IP addresses and upload it to the PAN UI.

Generate and import RSA configuration file

Follow these steps to generate and import the RSA configuration file:

1. Generate the **Authentication Manager** configuration file at your primary RSA Authentication Manager Security Console instance, with all the IP addresses of all the nodes, including those that are not in the deployment.
2. Import the new configuration file to PAN UI.

Note You must clear the node secret on your RSA Authentication Manager before uploading the new RSA configuration file. This helps to create a new node secret and share it between Cisco ISE and your RSA Authentication Manager.

3. Add a new node to the deployment without generating a new configuration file as it is replicated as part of the configuration using the IP addresses that are already present in the imported configuration file.

Process overview

This section presents upgrade steps for Cisco ISE when using the backup and restore method.

1. **Deregister a node**: To remove a node from the deployment, deregister the node. For more information about node deregistration or removal, see "Remove a Node from Deployment" section in the *Cisco ISE Administrator Guide, Release 3.3*.
2. **Reimage a node**: To reimage a Cisco ISE node, first remove it from the deployment, then install Cisco ISE. For more information about Cisco ISE installation, see the "Install Cisco ISE" chapter in *Cisco ISE Administrator Guide, Release 3.3*.

   Apply the latest patch to the newly installed Cisco ISE release.
3. **Back up and restore**: Back up and restore the configuration or operational database. For more information about the backup and restore operations, see the "Backup and Restore Operations" section in the *Cisco ISE Administrator Guide, Release 3.3*.
4. **Assign primary or secondary roles to a node**: Assign a primary or secondary role to a node as required. For more information about how to assign a role to a Monitoring and Troubleshooting (MnT) node, see the "Manually Modify MnT Role" section in the *Cisco ISE Administrator Guide, Release 3.3*.
5. **Join the Policy Service Nodes**: To join a Policy Service Node (PSN) to the new deployment, register the node as a PSN. For more information about registering or joining a PSN, see the "Register a Secondary Cisco ISE Node" section in the *Cisco ISE Administrator Guide, Release 3.3*.

Note After you upgrade Cisco ISE using the backup and restore method, you must manually sync all the nodes in the deployment.

6. **Import certificates**: Import the system certificates to the newly deployed nodes in Cisco ISE. For more information about how to import system certificates to a Cisco ISE node, see the "Import a System Certificate" section in the *Cisco ISE Administrator Guide, Release 3.3*.

Upgrade process

If you are currently using Cisco ISE release 3.0 or later, you can directly upgrade to Cisco ISE release 3.3.

In case you are using a Cisco ISE version that is not compatible with Cisco ISE release 3.3, you need to

1. firstly upgrade to an intermediate version, compatible with Cisco ISE release 3.3, and then
2. upgrade from the intermediate version to Cisco ISE release 3.3.

Follow these steps to upgrade to an intermediate Cisco ISE version.

Stage 1: Upgrade secondary PAN and secondary MnT nodes to Cisco ISE release 3.0, 3.1 or 3.2

Before you begin

Restore backup from your existing Cisco ISE to the intermediate Cisco ISE release. If you do not want to retain the older reporting data, skip steps 4 to 6.

Procedure

Step 1 Deregister secondary PAN node.

Step 2 Reimage the deregistered secondary PAN node to the intermediate Cisco ISE release, as a standalone node. After the installation, make this node the PAN in the new deployment.

Step 3 Restore Cisco ISE configuration from the backup data.

Step 4 Deregister secondary MnT node.

Step 5 Reimage the deregistered secondary MnT node to the intermediate Cisco ISE release as a standalone node.

Step 6 Assign primary role to this Mnt node and restore the operational backup from the backup repository. This is an optional step and needs to performed only if you need to report of the older logs.

Step 7 Import ise-https-admin CA certificates from your original Cisco ISE backup repository.

Stage 2: Upgrade secondary PAN and MnT nodes to Cisco ISE release 3.3

Procedure

Step 1 Back up your Cisco ISE configuration settings and operational logs.

Step 2 Deregister the secondary PAN node.

Step 3 Reimage your deregistered secondary PAN node to Cisco ISE release 3.3.

Step 4 Restore Cisco ISE configuration from the backup data and make this node the primary node for your new deployment.

Step 5 Import ise-https-admin CA Certificates from the backup for this node, unless you are using wildcard certificates.

Step 6 Deregister secondary MnT node.

Step 7 Reimage the deregistered Secondary MnT node to Cisco ISE release 3.3.

Step 8 Restore your current Cisco ISE operational backup and add the node as the primary MnT for your new deployment.

Perform this optional step only if you need to generate reports from the older logs.

Stage 3: Join PSNs to Cisco ISE release 3.3

If you have Cisco ISE nodes deployed in multiple sites, then you must

1. join the PSNs at the site containing secondary PAN and MnT nodes
2. join PSNs at the other sites, and then
3. join the PSNs at the site that has primary PAN and MnT nodes of your existing Cisco ISE.

Perform these steps to join PSNs to Cisco ISE release 3.3:

Procedure

Step 1 Deregister the PSNs.

Step 2 Reimage PSN to Cisco ISE release 3.3 latest patch and join PSN to new Cisco ISE release 3.3 deployment.

What to do next

We recommend that you test your partially upgraded deployment at this point. You can do so by checking that logs are present and that the upgraded nodes function as expected.

Stage 4: Upgrade Primary PAN and MnT to Cisco ISE release 3.3

Follow these steps to upgrade Primary PAN and MnT to the new Cisco ISE release:

Procedure

Step 1 Reimage the Primary MnT node and join it as the Secondary MnT to the new deployment.

To preserve data for reporting, restore an operational backup to the Secondary MnT node.

Step 2 Reimage the Primary PAN node and join it as the Secondary PAN to the new deployment.

Upgrade using the CLI

Overview

To upgrade Cisco ISE from the CLI, download the upgrade image to the local node, execute the upgrade, and monitor each node individually throughout the process. While the upgrade sequence is similar to that of the GUI upgrade, this approach is operationally intensive in terms of monitoring and actions.

Upgrading Cisco ISE from the CLI can be complex. Use CLI upgrades only for troubleshooting, as this method requires significant effort.

Advantages of upgrading from this method

The advantages of upgrading Cisco ISE from the CLI are:

- CLI presents additional logging messages to the administrator while the upgrade is performed.
- You can choose which nodes to upgrade with more control and upgrade them in parallel. Nodes not being upgraded handle additional load because endpoints are rebalanced across the deployment.
- You can easily roll back at the CLI because you can instruct scripts to undo previous changes.
- Because the image resides locally on the node, copy errors between the PAN and the PSNs are eliminated.

Key considerations before the upgrade

You should consider these points before upgrading Cisco ISE using the CLI method.

- Upgrading your Cisco ISE using CLI requires technical expertise and additional time.
- The upgrade process using CLI depends on the deployment type.

Upgrade a standalone node

To upgrade a standalone node, complete any one of these two steps:

- Enter this command directly:

  ```
  application upgrade <upgrade bundle name> <repository name>
  ```

- Enter these two commands in the specified order:

  ```
  application upgrade prepare <upgrade bundle name> <repository name>
  application upgrade proceed
  ```

You can run the **application upgrade <upgrade bundle name> <repository name>** command from the CLI on a standalone node that assumes the Administration, Policy Service, pxGrid, and Monitoring personas. Copy the upgrade bundle from the remote repository to the local disk of the Cisco ISE node before running the command. Saving the upgrade bundle to the local disk reduces the time required for the upgrade.

You can also upgrade a standalone node by completing these steps:

- Run this command to download the upgrade bundle and extract it locally:

  ```
  application upgrade prepare <upgrade bundle name> <repository name>
  ```

  This command copies the upgrade bundle from the remote repository to the local disk of Cisco ISE node.

- Run this command after preparing the node for upgrade to complete the process successfully:

  ```
  application upgrade proceed
  ```

Run the **application upgrade prepare <upgrade bundle name> <repository name>** and **application upgrade proceed** commands as described in this section.

Before you begin

Ensure that you have read the instructions in the "Prepare for Upgrade" chapter in this guide.

Procedure

Step 1 Enter this command in the CLI:

`application upgrade prepare <upgrade bundle name> <repository name>`

Step 2 Log in via SSH and use the **show application status ise** command to view the progress of the upgrade.

This message appears: % NOTICE: Identity Services Engine upgrade is in progress...

Step 3 Enter the **application upgrade proceed** command.

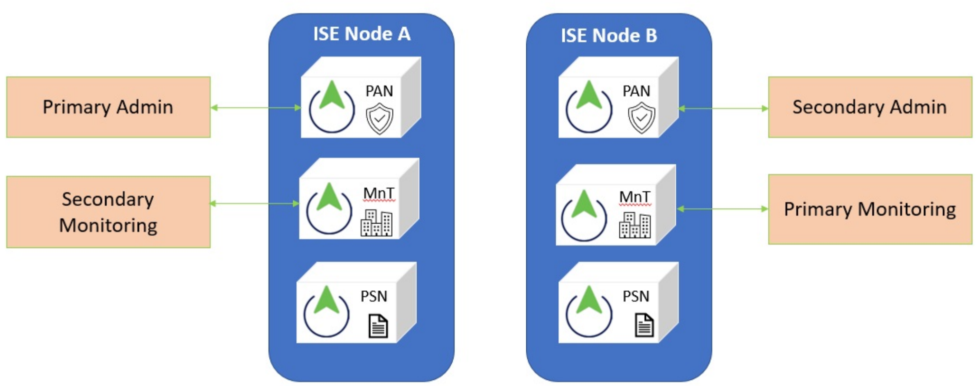
Upgrade a two-node deployment

Use these two commands to upgrade a two-node deployment:

1. `application upgrade prepare <upgrade bundle name> <repository name>`
2. `proceed`

You do not have to manually deregister the node and register it again. The upgrade software automatically deregisters the node and adds it to the new deployment. When you upgrade a two-node deployment, begin by upgrading only the SAN (node B). After the secondary node upgrade is complete, upgrade the primary node (node A). If you have a deployment set up as shown in this figure, you can proceed with this upgrade procedure.

Figure 1: Cisco ISE two-node administrative deployment



Before you begin

- Perform a manual backup of the configuration and operational data from the PAN.
- Enable the Administration and Monitoring personas on both nodes in your deployment.

If only the PAN has the Administration persona enabled, enable the Administration persona on the secondary node. You must upgrade the SAN first.

If your two-node deployment has only one Administration node, deregister the secondary node. Each node operates as a standalone node. Upgrade each standalone node, then set up your deployment again after the upgrade.

- If only one node has the Monitoring persona enabled, enable the Monitoring persona on the other node before you continue.

Procedure

Step 1 Upgrade the secondary node (node B) from the CLI.

The upgrade process automatically removes Node B from deployment and upgrades it. Node B becomes the upgraded primary node when it restarts.

Step 2 Upgrade node A.

The upgrade process automatically registers node A to the deployment and makes it the secondary node in the upgraded environment.

Step 3 Promote node A, now to primary node in the new deployment.

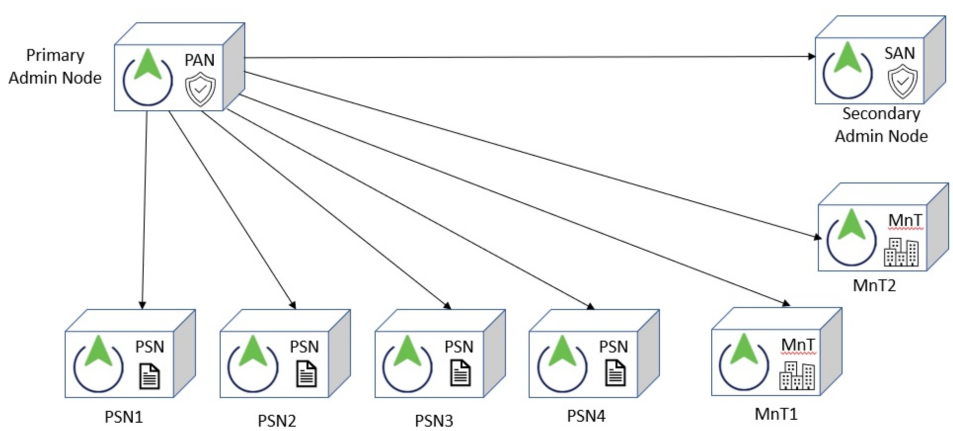
After the upgrade is complete, if the nodes contain old Monitoring logs, run this command and choose 5 (Refresh Database Statistics) on the nodes.

```
application configure ise
```

Upgrade a distributed deployment

You must first upgrade the SAN to the new release. For example, if you have a deployment setup as shown in this figure, with one PAN, one SAN, four PSNs, one Primary Monitoring Node (MnT1), and one Secondary Monitoring Node (MnT2), you can proceed with this upgrade procedure.

Figure 2: Cisco ISE deployment before upgrade



Note Do not manually deregister the node before an upgrade. Use these commands to upgrade to the new release:

1. application upgrade prepare <upgrade bundle name> <repository name>
2. proceed

The upgrade process deregisters the node automatically and moves it to the new deployment. If you manually deregister the node before upgrading, ensure you have the PAN license file before starting the upgrade. If you do not have the file (such as when a Cisco partner vendor installed your license), contact the Cisco Technical Assistance Center.

Follow these steps for a smooth upgrade while maintaining deployment integrity and minimizing downtime.

Before you begin

- If you do not have a SAN in the deployment, configure a PSN to be the SAN before beginning the upgrade process.
- Ensure that you have read and complied with the instructions given in the Prepare for Upgrade chapter of this guide (*Cisco ISE Upgrade Journey, Release 3.3*).
- When you upgrade a complete Cisco ISE deployment, Domain Name System (DNS) server resolution (both forward and reverse lookups) is mandatory; otherwise, the upgrade fails.

Procedure

Step 1 **Upgrade the SAN from the CLI**

- Initiate the upgrade from the CLI on the SAN node.
- The upgrade process automatically deregisters SAN from the old deployment and upgrades it.
- Upon restart, SAN becomes the primary node of the new deployment.
- Each deployment requires at least one Monitoring node; therefore, the upgrade enables the Monitoring persona on SAN, even if it was not previously enabled.
- If the Policy Service persona was enabled on SAN in the old deployment, it is retained after upgrade.

Step 2 **Upgrade a Monitoring node (MnT1 and MnT2)**

- Upgrade one Monitoring node to the new deployment.
- If the PAN in the old deployment also serves as the Primary Monitoring node, you cannot upgrade the Primary Monitoring node before the Secondary Monitoring node. Otherwise, upgrade the Primary Monitoring node first.
- The Primary Monitoring node begins collecting logs from the new deployment, accessible via the PAN dashboard.
- If only one Monitoring node exists in the old deployment, enable the Monitoring persona on PAN before upgrading it.

Note

Changing node personas causes a Cisco ISE application restart; wait for the node to come back online before proceeding.

- Upgrading the Monitoring node takes longer due to operational data migration.
- If the PAN in the new deployment did not have the Monitoring persona enabled previously, disable it on that node, which also triggers an application restart.

Step 3 **Upgrade PSNs**

- Next, upgrade the PSNs.
- Multiple PSNs can be upgraded in parallel, but upgrading all simultaneously may cause network downtime.
- After upgrade, PSNs register with the new deployment's primary node (SAN).
- PSNs retain their personas, node group assignments, and profiling probe configurations.

Step 4 **Upgrade the second Monitoring node (if applicable)**

- If a second Monitoring node exists in the old deployment:
  - Enable the Monitoring persona on PAN (primary node in the old deployment). This is required as each deployment must have at least one Monitoring node. Persona changes cause an application restart; wait for the node to come back online.
  - Upgrade the Secondary Monitoring node to the new deployment.
  - Ensure all nodes except the PAN have been upgraded before this step.

Step 5 **Upgrade the PAN**

- Upgrade the PAN last.
- After upgrade, PAN joins the new deployment as a SAN.
- You can promote this SAN to Primary node if needed.
- Post-upgrade, if the Monitoring nodes contain old logs, run this command:

  application configure ise
- Select option 5 (Refresh Database Statistics) on those nodes.

Roll back to the previous version

In rare cases where you need to reimage your Cisco ISE appliance, you should use the previous ISO image and restore the data from the backup file. After restoring the data, you can register the appliance with the earlier deployment. Enable the personas as they were in the previous deployment. It is important to back up the Cisco ISE configuration and monitoring data before starting the upgrade process.

Upgrade failures sometimes occur due to issues in the configuration and monitoring database. In these cases, the system does not roll back automatically. When this happens, the system displays a notification indicating that the database was not rolled back and provides an upgrade failure message. You must then manually reimage your system, install Cisco ISE, and restore the configuration data and and monitoring data if the Monitoring persona is enabled..

Before performing rollback or recovery, generate a support bundle using the **backup-logs** command and save it in a remote repository. This support bundle can be used for troubleshooting and submitted to Cisco Technical Assistance Center (TAC) if needed.

Verify the upgrade process

After completing the upgrade, it is important to verify that your deployment is functioning correctly. Run network tests to confirm that you can authenticate and access network resources as expected.

If the upgrade fails due to configuration database issues, the system will automatically roll back the changes.

Use these options to verify that the upgrade was successful.

Procedure

Step 1 Check the upgrade progress by reviewing the ade.log file. Use this command from the Cisco ISE CLI to display the log:

```
show logging system ade/ADE.log.?
```

You can filter the log for upgrade steps with the help of the keyword STEP. The log entries will show progress similar to the following:

- `info:[application:install:upgrade:preinstall.sh] STEP 0: Running pre-checks`
- `info:[application:operation:preinstall.sh] STEP 1: Stopping ISE application...`
- `info:[application:operation:preinstall.sh] STEP 2: Verifying files in bundle...`
- `info:[application:operation:isedbupgrade-newmodel.sh] STEP 3: Validating data before upgrade...`
- `info:[application:operation:isedbupgrade-newmodel.sh] STEP 4: De-registering node from current deployment.`
- `info:[application:operation:isedbupgrade-newmodel.sh] STEP 5: Taking backup of the configuration data...`
- `info:[application:operation:isedbupgrade-newmodel.sh] STEP 6: Registering this node to primary of new deployment...`
- `info:[application:operation:isedbupgrade-newmodel.sh] STEP 7: Downloading configuration data from primary  of new deployment...`
- `info:[application:operation:isedbupgrade-newmodel.sh] STEP 8: Importing configuration data...`
- `info:[application:operation:isedbupgrade-newmodel.sh] STEP 9: Running ISE configuration data upgrade for node specific data...`
- `info:[application:operation:isedbupgrade-newmodel.sh] STEP 10: Running ISE M&T database upgrade...`
- `info:[application:install:upgrade:post-osupgrade.sh] POST ADEOS UPGRADE STEP 1: Upgrading Identity Services Engine software...`
- `info:[application:operation:post-osupgrade.sh] POST ADEOS UPGRADE STEP 2: Importing upgraded data to 64 bit database...`

- Search the log for this string to confirm a successful upgrade:

```
Upgrade of Identity Services Engine completed successfully.
```

Step 2 Verify the build version by entering this command:

show version

Step 3 Confirm that all Cisco ISE services are running by entering this command:

show application status ise

Verify patch installation on nodes

If you have opted for the upgrade process and patch installation, it is important to verify that patches are correctly installed on all Cisco ISE nodes. Ensuring that all nodes run the correct and consistent patch level across the deployment is essential to maintain system stability and prevent issues such as replication failures and service disruptions.

Procedure

Step 1 In the Cisco ISE GUI, navigate to **Administration** > **System** > **Maintenance** > **Patch Management** .

Step 2 Locate the patch in the list and click the radio button next to it.

Step 3 Click **Show Node Status**.

Step 4 A dialog box will appear displaying the status of each node. Confirm that every node is listed as **Installed**.

If a node does not have the patch installed and shows the status as **Uninstalled**, follow the CLI patch installation process to install the patch on that node.

Cisco ISE installs patches starting from the PAN and then proceeds to secondary nodes. If patch installation fails on the PAN, it does not proceed to secondary nodes. However, if installation fails on any secondary node, the process continues with the next node. This ensures that patch installation is managed consistently across the deployment.

Maintaining consistent patch levels is critical because inconsistent patching can cause severe system instability. Therefore, verification of patch installation on all nodes after an upgrade or patch installation is a mandatory step to ensure system health and stability.

CHAPTER 4

Install Latest Patch

Cisco ISE software patches

Cisco ISE software patches are always cumulative. You can perform patch installation and rollback using these options:

- Patch installation from Primary PAN: Patches are installed on Cisco ISE servers in your deployment starting from the Primary PAN. To install a patch from the Primary PAN, download the patch file from Cisco.com to the system running your client browser.
- Patch installation using the GUI: When installing a patch using the GUI, the system installs the patch on the Primary PAN first. It then installs the patch on the remaining nodes in the deployment following the order displayed in the GUI, which cannot be changed. You can also manually install patches, roll back patches, and view patch versions by navigating to this path in the Cisco ISE GUI:

  Administrator > **System** > **Maintenance** > **Patch Management**
- Using the CLI: Installing patches from the CLI allows you to control the update order of nodes. It is recommended to install the patch on the Primary PAN first, but the order for other nodes is flexible. You can install patches on multiple nodes simultaneously to expedite the process. To install a patch on specific nodes for validation before upgrading the entire deployment, use the CLI command:

  ```
  patch install <patch_bundle> <repository_that_stores_patch_file>
  ```

For more information, see "Install Patch" in the "Cisco ISE CLI Commands in EXEC Mode" chapter in the *Cisco ISE CLI Reference Guide, Release 3.3*.

You can install the required patch version directly. For example, if you are using Cisco ISE release 3.x and want to install patch 5, you can install patch 5 without installing patches 1 through 4.

To view the current patch version in the CLI, use this command:

show version

Software patch installation guidelines

Follow these guidelines while installing software patches:

- When you install a patch on a Cisco ISE node, the node will reboot after the installation completes. You may need to wait a few minutes before you can log in again. Schedule patch installations during maintenance windows to minimize service disruption.
- Ensure that the patch you install is compatible with the Cisco ISE version deployed in your network. Cisco ISE will report any version mismatches or errors in the patch file.
- You cannot install a patch with a version lower than the currently installed patch on Cisco ISE. Similarly, rolling back to a lower-version patch is not allowed if a higher version is installed. For example, if patch 3 is installed, you cannot install or roll back to patch 1 or 2.
- In a distributed deployment, when installing a patch from the Primary PAN, Cisco ISE installs the patch on the primary node first, then proceeds to the secondary nodes. If the patch installation succeeds on the Primary PAN, the process continues on the secondary nodes. If it fails on the Primary PAN, installation does not proceed to secondary nodes. If installation fails on any secondary node, the process continues with the next secondary node.
- In a two-node deployment, Cisco installs the patch from the Primary PAN on the primary node first and then on the secondary node. If installation fails on the Primary PAN, it does not proceed to the secondary node.

Install a software patch

Before you begin

- You must be assigned the Super Admin or System Admin role.
- The PAN auto-failover configuration must be disabled for the duration of this task.

  To disable this setting, complete these steps:

  1. In the Cisco ISE GUI, click the **Menu** icon (☰) and choose **Administration** > **System** > **Deployment** > **PAN Failover**.
  2. Uncheck the **Enable PAN Auto Failover** check box.

Procedure

Step 1 In the Cisco ISE GUI, click the **Menu** icon (☰) and choose **Administration** > **System** > **Maintenance** > **Patch Management** > **Install.**

Step 2 Click **Browse** and choose the patch that you downloaded from Cisco.com.

Step 3 Click **Install** to install the patch.

After the patch is installed on the PAN, Cisco ISE logs you out. You must wait a few minutes before logging in again.

When patch installation is in progress, **Show Node Status** is the only function that is accessible on the Patch Management page.

Step 4 In the Cisco ISE GUI, click the **Menu** icon (☰) and choose **Administration** > **System** > **Maintenance** > **Patch Management** to return to the Patch Installation page.

Step 5 Click the radio button next to the patch that you have installed. Click **Show Node Status** to verify installation is complete.

Verify patch installation

It is mandatory to ensure that the patch is installed on all nodes within your deployment. Inconsistent patch levels across nodes can cause severe system instability, including replication failures and service disruptions. The upgrade process is not considered complete until every node reflects the correct patch version.

To verify that the patch is correctly installed on all nodes, follow these steps:

Procedure

Step 1 In the Cisco ISE GUI, navigate to **Administration** > **System** > **Maintenance** > **Patch Management** .

Step 2 Locate the patch in the list and click the radio button next to it.

Step 3 Click **Show Node Status**.

Step 4 A dialog box will appear displaying the status of each node. Confirm that every node is listed as **Installed**.

If a node does not have the patch installed and shows the status as **Uninstalled**, follow the CLI patch installation process to install the patch on that node.

Roll back software patches

When you roll back a patch from the PAN in a deployment with multiple nodes, Cisco ISE rolls back the patch on the primary node and then on all the secondary nodes, depending on the deployment.

Before you begin

- You must be assigned the Super Admin or System Admin role.

Procedure

Step 1 In the Cisco ISE GUI, navigate to **Administration** > **System** > **Maintenance** > **Patch Management**.

Step 2 Select the patch version to roll back and then click **Rollback**.

When a patch rollback is in progress, only the **Show Node Status** function is accessible on the Patch Management page.

Cisco ISE logs you out after the patch is rolled back from the PAN. Wait a few minutes before you log in again.

Step 3 After you log in, click the **Alarms** link at the bottom of the page to view the status of the rollback operation.

Step 4 To view the progress of the patch rollback, choose the patch on the Patch Management page and click **Show Node Status**.

Step 5 Select the patch and click **Show Node Status** on a secondary node to ensure the patch is rolled back from all nodes in your deployment.

If the patch is not rolled back from any secondary node, ensure the node is operational. Repeat this process to roll back changes from any remaining nodes. Cisco ISE rolls back the patch only from nodes that still have this version of the patch installed.

Software patch rollback guidelines

To roll back a patch from Cisco ISE nodes in a deployment, you must do the following:

- Roll back the patch first from PAN.
- If the rollback on the PAN is successful, roll back the patch from the secondary nodes.
- If the rollback fails on the PAN, do not roll back the patches from the secondary nodes.
- If the rollback fails on any secondary node, continue to roll back the patch on the next secondary node in the deployment.

While Cisco ISE rolls back the patch from the secondary nodes, you can continue to perform other tasks from the PAN GUI. The secondary nodes will be restarted after the rollback.

View patch install and roll back changes

You can install or roll back patches from the Cisco ISE GUI. In the Cisco ISE GUI, click the **Menu** icon () and choose **Administration** > **System** > **Maintenance** > **Patch Management** window. You can view the status for each node (installed, in progress, or not installed) by selecting a patch and clicking **Show Node Status**.

To view reports for installed patches, perform these steps:

Before you begin

You must be assigned the Super Admin or System Admin role.

Procedure

Step 1 In the Cisco ISE GUI, click the **Menu** icon () and choose **Operations** > **Reports** > **Audit** > **Operations Audit**.

By default, records for the last seven days are displayed.

Step 2 Click the **Filter** drop-down and choose **Quick Filter** or **Advanced Filter**. Enter the required keyword (for example, 'patch install initiated') to generate a report of installed patches.

CHAPTER 5

Perform the Postupgrade Tasks

After you upgrade your deployment, perform the tasks listed in this chapter.

Postupgrade settings and configurations

Perform these tasks after upgrading Cisco ISE.

Convert to new license types

To convert to new license types, perform these steps:

1. Convert your old licenses to the new license types through the Cisco Smart Software Manager (CSSM).
2. Enable the new licenses in your Cisco ISE administrator portal.

For more information, see "Cisco ISE Licenses" in the "Licensing" chapter in the *Cisco ISE Administrator Guide, Release 3.3*.

Verify VM settings

If you are upgrading Cisco ISE nodes on VMs, change the Guest OS to RHEL 8.4 (64-bit). To do this

1. power down the VM
2. change the Guest OS to the supported RHEL version, and then
3. power on the VM.

RHEL 7 and later support only E1000 and VMXNET3 network adapters. Change your network adapter type before you upgrade.

Browser setup

After you upgrade, you must

1. clear your browser cache

2. close your browser, and
3. open a new browser session (on a supported browser) to access the Cisco ISE Admin portal.

> **Note** See the release notes for a list of supported browsers:https://www.cisco.com/c/en/us/support/security/identity-services-engine/products-release-notes-list.html.

Rejoin Active Directory

If you log in to the Cisco ISE user interface with an AD administrator account after the upgrade, the login fails because the AD join is lost during the upgrade.

If you use certificate-based authentication for administrative access and AD is your identity source, you cannot access the login page after an upgrade. This is because the AD join is lost during the upgrade.

If you use Active Directory (AD) as your external identity source and lose connection, complete these steps:

1. Enter this command to start Cisco ISE in safe mode:

   application start ise safe

2. Log in to the Cisco ISE user interface using the internal administrator account.

> **Note** If you forgot your password or your administrator account is locked, see the "Administrator Access to Cisco ISE" chapter in the *Cisco ISE Administrator Guide, Release 3.3*.

3. Rejoin Cisco ISE with AD.
4. Run the external identity source workflows to verify that the connection is restored.

Certificate attributes with AD

Cisco ISE identifies users by their attributes *SAM*, *CN*, or both, with *sAMAccountName* attribute used as default.

You can configure Cisco ISE to use *SAM*, *CN*, or both, depending on your environment. If both SAM and CN are used and the *sAMAccountName* attribute is not unique, Cisco ISE also compares the CN attribute value.

To configure attributes for AD identity search, complete these steps:

1. In the Cisco ISE GUI, navigate to **Administration** > **Identity Management** > **External Identity Sources** > **Active Directory**.
2. Click **Advanced Tools**.
3. Choose **Advanced Tuning** and enter these details:
   - **ISE Node**: Choose the Cisco ISE node that is connecting to Active Directory.

- **Name**: Enter the registry key that you are changing. To change the Active Directory search attributes, enter: `REGISTRY.Services\lsass\Parameters\Providers\ActiveDirectory\IdentityLookupField`
- **Value**: Enter the attributes that Cisco ISE uses to identify a user:
  - *SAM*: To use only SAM in the query (this option is the default).
  - *CN*: To use only CN in the query.
  - *SAMCN*: To use CN and SAM in the query.
- **Comment**: Describe what you are changing, for example: Changing the default behavior to SAM and CN.

4. Click **Update Value** to update the registry.

   When the pop-up window appears, read the message and accept the change. The AD connector service in Cisco ISE restarts automatically.

Reverse DNS lookup

Configure Reverse DNS lookup for all Cisco ISE nodes in your distributed deployment on every DNS server. If you do not configure reverse DNS lookup, deployment-related issues may occur after the upgrade.

Restore certificates

This section details how to restore certificates and keys on Cisco ISE Administration Nodes to prevent authentication failures that may occur during upgrades.

Restore certificates on the PAN

When you upgrade a distributed deployment, the PAN's root CA certificates are not added to the Trusted Certificates store if both of these conditions are met:

- SAN is promoted to be the PAN in the new deployment.
- Session services are disabled on the SAN.

If the certificates are not in the store, you may see authentication failures with these errors:

- `Unknown CA in the chain during a BYOD flow`
- `OCSP unknown error during a BYOD flow`

You can see these messages when you click the **More Details** link from the **Live Logs** page for failed authentications.

To restore the root CA certificates of PAN, generate a new Cisco ISE Root CA certificate chain. In the Cisco ISE GUI, navigate to **Administration** > **Certificates** > **Certificate Signing Requests** > **Replace ISE Root CA certificate chain**.

Restore certificates and keys to SAN

If you are using a SAN, obtain a backup of the Cisco ISE CA certificates and keys from the PAN, and restore it on the SAN. This allows the SAN to function as the root CA or subordinate CA of an external PKI if the primary PAN fails, and you promote the SAN to be the PAN.

For more information, see "Backup and restoration of Cisco ISE CA certificates and keys" in the "Basic Setup" chapter in the *Cisco ISE Administrator Guide, Release 3.3*.

Regenerate the root CA chain

If your deployment matches specific upgrade scenarios, you must regenerate the root CA chain after the upgrade is complete.

To regenerate the root CA chain, complete these steps:

1. In the Cisco ISE GUI, navigate to **Administration** > **System** > **Certificates** > **Certificate Management** > **Certificate Signing Request**.
2. Click **Generate Certificate Signing Request (CSR)**.
3. Choose **ISE Root CA** in the **Certificate(s) will be used for** drop-down list.
4. Click **Replace ISE root CA Certificate Chain**.

This table defines various root CA chain regeneration scenarios.

Table 6: Root CA chain regeneration scenarios

Upgrade scenario	Mode	F
Full upgrade process	Deployment and Standalone	Y
Split upgrade process	Deployment and Standalone	T
Configuration database restoration process	Standalone	T
Node Promotion: Promoting a secondary PAN to primary PAN after the split upgrade process	Deployment	F
Change in the domain name or hostname of any Cisco ISE node	Standalone and Deployment	F

After the upgrade process, you might encounter these events:

- Data might not be available in live logs.
- You might see queue link errors.
- The system might show the health status as unavailable.
- System summary might not display data for some nodes.

To resolve queue link errors and restore system information,

- reset the MnT Database , and
- replace the ISE Root CA certificate chain.

Threat-centric NAC

If you enable the Threat-centric NAC (TC-NAC) service, the TC-NAC adapters might not function after an upgrade. Restart the adapters from the TC-NAC pages of the Cisco ISE GUI. Select an adapter and click **Restart**.

Set the SNMP originating policy services node

If you manually configure the SNMP originating policy services node setting value under SNMP settings, you lose the configuration during an upgrade. Reconfigure the SNMP settings to restore the SNMP functionality.

Profiler feed service

After you upgrade, update the profiler feed service to ensure that the most up-to-date Organizationally Unique Identifiers (OUIs) are installed.

Follow these steps to update the profiler feed service:

Procedure

Step 1 In the Cisco ISE GUI, navigate to **Administration** > **FeedService** > **Profiler**.

Step 2 Ensure the profiler feed service is enabled.

Step 3 Click **Update Now**.

Client provisioning

Review these considerations before updating client provisioning resources:

- Check the native supplicant profile used in the client provisioning policy.
- Ensure that the wireless SSID is correct.
- For iOS devices, if the network you are trying to connect to is hidden, check the **Enable if target network is hidden** check box in the **iOS Settings** area.

Follow these steps to update client provisioning resources:

Online updates

Procedure

Step 1 In the Cisco ISE GUI, navigate to **Policy > Policy Elements > Results > Client Provisioning > Resources** to configure the client provisioning resources.

Step 2 Click **Add**.

Step 3 Choose **Agent Resources From Cisco Site**.

Step 4 In the **Download Remote Resources** window, select the Cisco Temporal Agent resource.

Step 5 Click **Save** and verify that the downloaded resource appears in the Resources page.

Offline updates

Procedure

Step 1 In the Cisco ISE GUI, navigate to **Policy > Policy Elements > Results > Client Provisioning > Resources** to configure the client provisioning resources.

Step 2 Click **Add** to add a new resource.

Step 3 Choose **Agent Resources from Local Disk**.

Step 4 From the **Category** drop-down list, choose **Cisco Provided Packages**.

Cipher suites

Authentication fails for legacy devices that use deprecated ciphers when connecting to Cisco ISE.

To allow Cisco ISE to authenticate legacy devices after upgrading, update the **Allowed Protocols** configuration by following these steps:

Procedure

Step 1 In the Cisco ISE GUI, navigate to **Policy** > **Policy Elements** > **Results** > **Authentication** > **Allowed Protocols**.

Step 2 Edit the **Allowed Protocols** service and check the **Allow weak ciphers for EAP** check box.

Step 3 Click **Submit**.

Monitor and troubleshoot

Consider these steps to monitor and troubleshoot:

- Reconfigure your email settings.
- Update your favorite reports.
- Change your data purge settings.
- Check thresholds and filters for specific alarms you need.

Note By default, all alarms are enabled after an upgrade.

- Customize reports based on your needs.

> **Note** If you customized reports during the previous deployment, your changes will be replaced during the upgrade.

Restore the MnT backup that you created before the update.

Refresh policies to Trustsec NADs

Run these commands to download the policies to Cisco TrustSec-enabled Layer 3 interfaces in your system:

1. `no cts role-based enforcement`
2. `cts role-based enforcement`

Update Supplicant Provisioning Wizards

The Supplicant Provisioning Wizards (SPWs) are not updated when you upgrade to a new release or apply a patch. You must manually update the SPWs. Then, create new native supplicant profiles and new client provisioning policies that reference the new SPWs. You can find new SPWs on the Cisco ISE Download page. Visit the Cisco software download site for more information.

Profiler endpoint ownership synchronization or replication

During an upgrade, the JEDIS framework requires port 6379 to be open between all nodes in the deployment to allow two-way communication.